Introduction to Flip-Flops and Their Types

Subject: Digital Electronics and Computer Organization

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Project Topic: Flip-Flops

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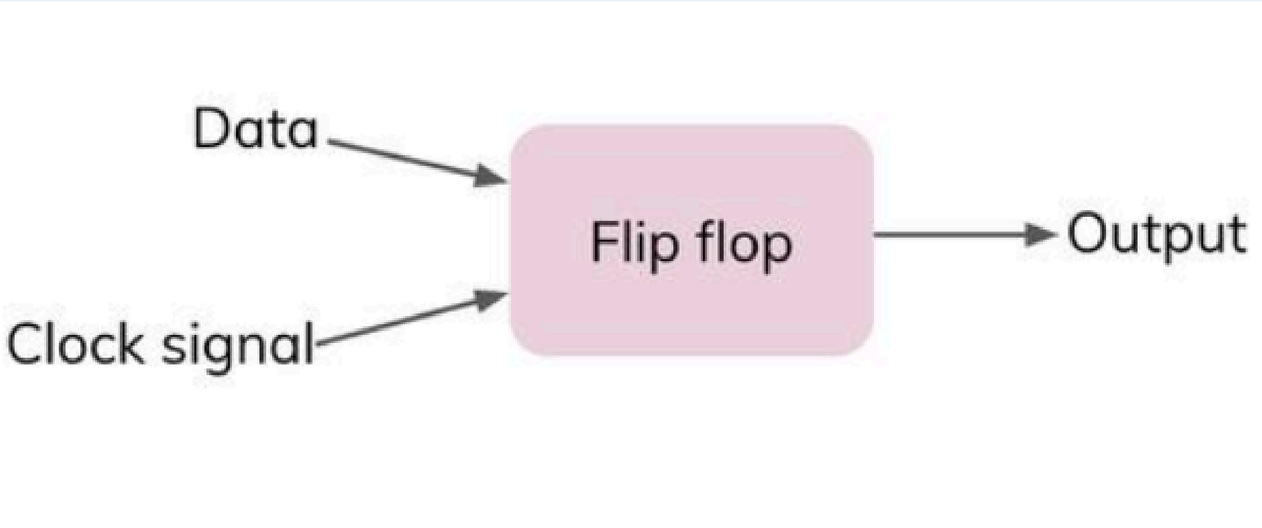
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Introduction to Flip-Flops

- Flip-flops are basic memory elements used in digital electronics.
- They are used to store binary information (0 or 1).
- A flip-flop has two stable states.
- It is an important part of sequential logic circuits.
- Flip-flops are widely used in computers, counters and memory devices.



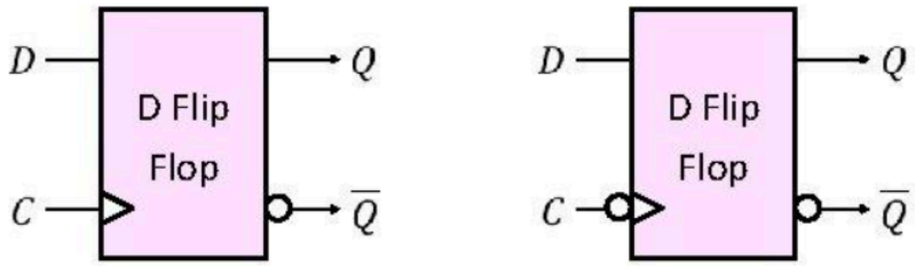
Characteristics of Flip-Flops

1. A flip-flop stores one bit of data.
2. It has two stable states: 0 and 1.
3. It works with a clock signal (in clocked types).
4. It is used in sequential circuits.
5. It can change its output according to input conditions.

Important Terms:

- Q = Normal Output
- $\bar{Q}$ = Complementary Output
- CLK = Clock Signal

Graphic Symbols for Flip-Flops



- A Flip-Flop has a similar symbol to a Latch
- The difference is the arrowhead at the clock input *C*
- The arrowhead indicates sensitivity to the edge of the clock
- A bubble at the *C* input indicates negative edge-triggered FF

Types of Flip-Flops

There are four main types of flip-flops:

1. SR Flip-Flop (Set-Reset)
2. JK Flip-Flop
3. D Flip-Flop (Data)
4. T Flip-Flop (Toggle)

Each flip-flop has different inpaut conditions and working operations.

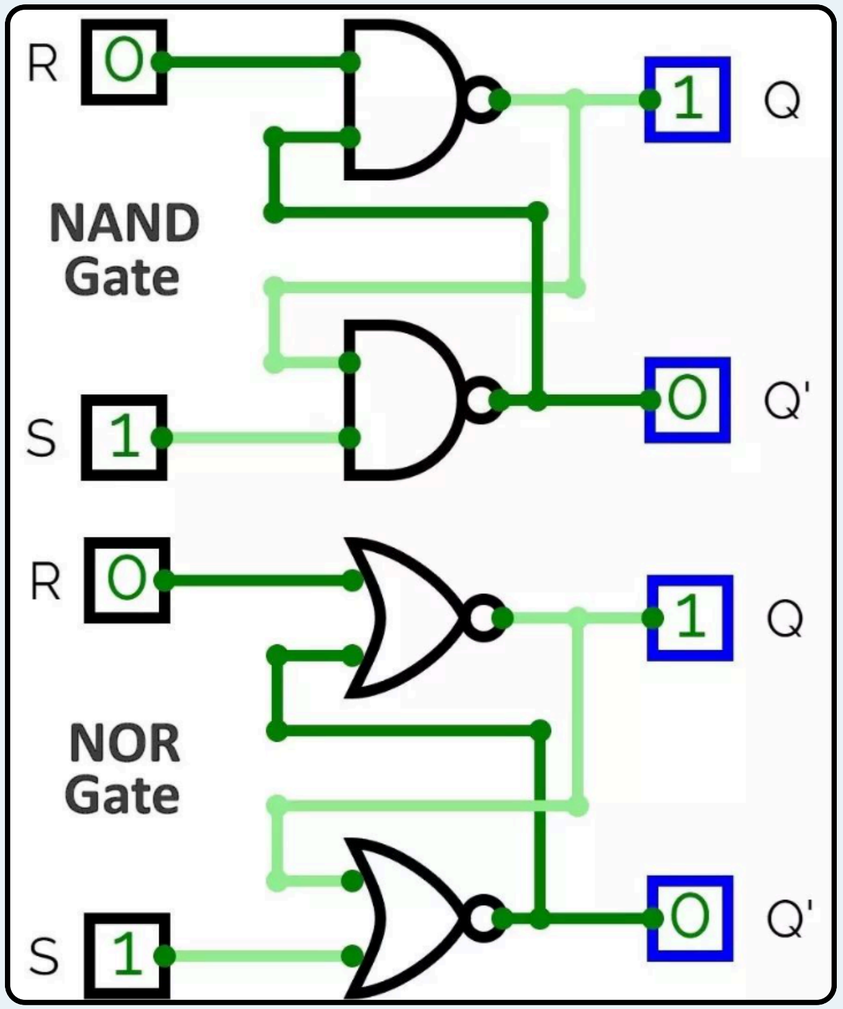
SR Flip-Flop

SR stands for Set-Reset Flip-Flop.

- It has two inputs: S (Set) and R (Reset).
- It has two outputs: Q and $\bar{Q}$.
- It is used to store one bit of information.
- It can set or reset the output.

Truth Table:

S	R	Q (Next State)
0	0	No Change
0	1	0 (Reset)
1	0	1 (Set)
1	1	Invalid

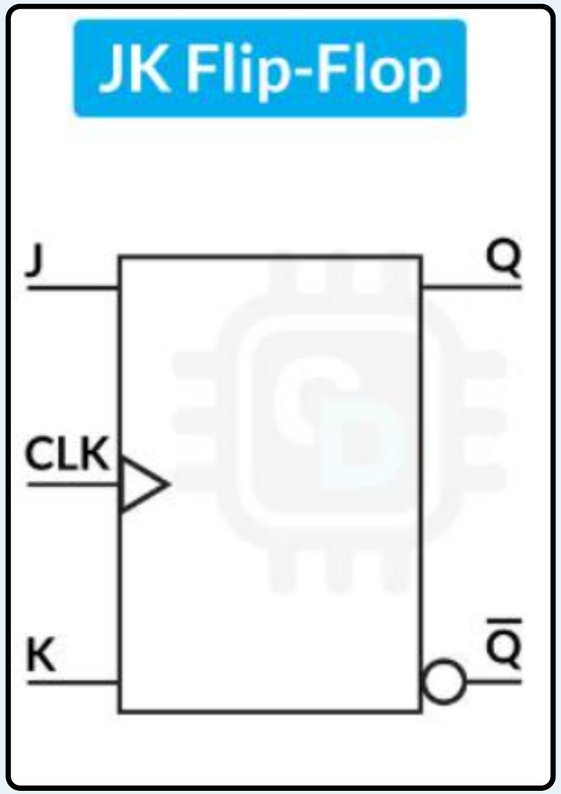


JK Flip-Flop

- JK Flip-Flop is an improved version of SR Flip-Flop.
- It has two inputs: J and K.
- It has two outputs: Q and $\bar{Q}$.
- It removes the invalid condition of the basic SR flip-flop.
- It is commonly used in counters and sequential circuits.

Truth Table:

J	K	Q (Next State)
0	0	No Change
0	1	Reset
1	0	Set
1	1	Toggle



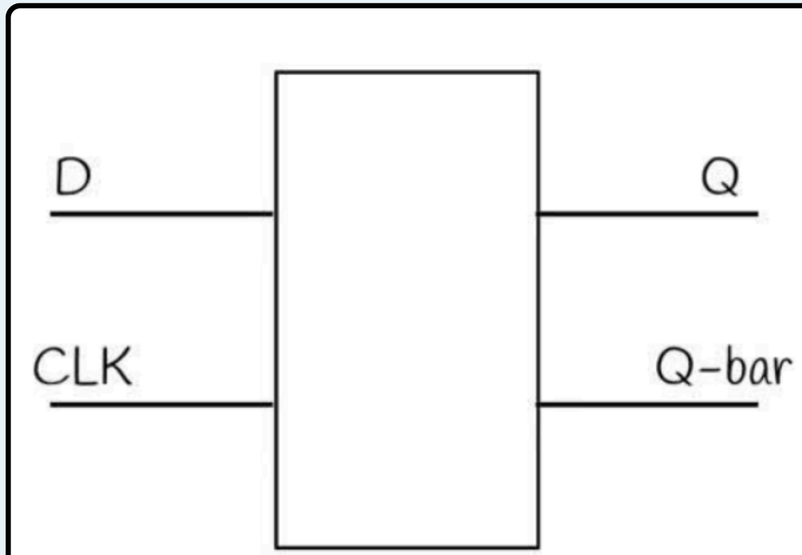
D Flip-Flop

D stands for Data Flip-Flop.

- It has only one main input called D.
- It stores the input data.
- The output follows the D input at the active clock edge.
- It is widely used in registers and memory circuits.
- It is also called a Delay Flip-Flop.

Truth Table:

D	Q (Next State)
0	0
1	1



CLK	D	Q
0	0	Q
0	1	Q
1	0	0
1	1	1

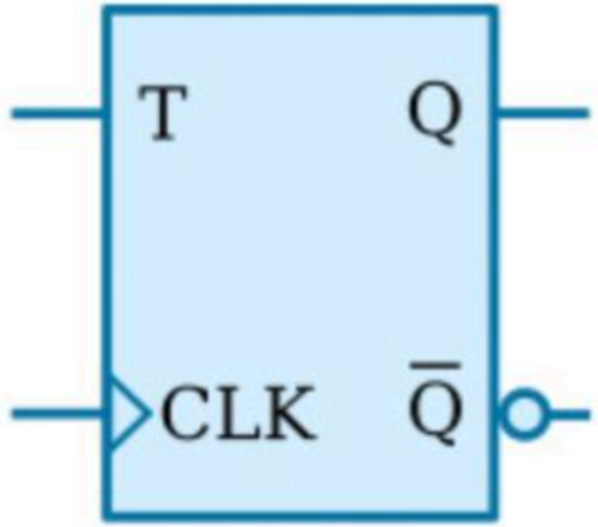
T Flip-Flop

T stands for Toggle Flip-Flop.

- It has one input called T.
- When T = 0, the output remains unchanged.
- When T = 1, the output toggles.
- It is mainly used in binary counters.
- It helps in frequency division circuits.

Truth Table:

T	Q (Next State)
0	No Change
1	Toggle



T	Q	Status
0	Q	Hold
1	Q'	Toggle

Applications of Flip-Flops

Flip-flops are used in many digital electronic devices.

1. Registers – To store binary data.
2. Counters – To count digital pulses.
3. Memory Units – To store information.
4. Frequency Dividers – To divide clock frequency.
5. Digital Systems – Used in computers and processors.

Conclusion

- Flip-flops are important memory elements in digital electronics.
- They store binary data in the form of 0 and 1.
- There are four main types: SR, JK, D and T.
- Each type performs a different operation.
- Flip-flops are essential components of modern digital systems.